

AI Healthcare Assistant using Hugging Face and Prompt Engineering

Introduction

The objective of this project was to develop an AI-powered healthcare assistant using the Hugging Face Transformers library and Prompt Engineering techniques. The chatbot is designed to answer medical questions, generate structured responses, and provide educational healthcare information. The project demonstrates how carefully designed prompts can improve the quality, consistency, and safety of responses generated by Large Language Models (LLMs).

The chatbot was implemented using Hugging Face's Transformers library with an instruction-tuned language model. Different prompt engineering techniques such as Role Prompting, Few-Shot Prompting, Chain-of-Thought Prompting, and Structured Output Prompting were applied to improve the chatbot's performance.

Model Selection

For this project, the **Qwen/Qwen2.5-1.5B-Instruct** model from Hugging Face was selected.

Reasons for Selecting the Model

- It is an instruction-tuned Large Language Model designed for conversational tasks.
- It generates high-quality and context-aware responses.
- It follows user instructions more accurately than base language models.
- It supports prompt engineering techniques effectively.
- It can be integrated easily using the Hugging Face Transformers library with AutoTokenizer, AutoModelForCausalLM, and the pipeline API.

Although the model requires more computational resources than smaller models, it produces significantly better responses for healthcare-related educational queries.

Prompt Engineering Techniques Used

Several prompt engineering techniques were applied throughout the project to improve the chatbot's performance.

1. Role Prompting

The model was assigned the role of an **AI Healthcare Assistant**. This encouraged the model to provide professional, informative, and medically appropriate responses while avoiding unnecessary or unrelated information.

2. Context Prompting

Each prompt included relevant patient information and clearly described the task. Providing context helped the model generate responses that were more relevant and consistent.

3. Constraint Prompting

Specific constraints were added to reduce hallucinations and improve reliability. Examples of constraints included:

- Do not guess medical diagnoses.
- Use only the provided information.
- Recommend consulting a healthcare professional when necessary.
- Avoid unnecessary medical jargon for patient-facing responses.

These constraints helped improve the safety and usefulness of the generated responses.

4. Few-Shot Prompting

Few-shot prompting was used by providing multiple examples of clinical notes along with their corresponding structured summaries. This allowed the model to learn the expected output format before processing new clinical notes.

5. Chain-of-Thought Prompting

The prompts encouraged the model to analyze patient symptoms carefully before generating recommendations. However, the prompt explicitly instructed the model **not to reveal its internal reasoning**, ensuring compliance with the assignment requirements while still improving response quality.

6. Structured Output Prompting

The chatbot was instructed to generate responses in JSON format for certain tasks. Structured outputs make it easier for applications to process AI-generated information and integrate it into healthcare systems.

Challenges Faced

Several challenges were encountered during the implementation of the project.

- Loading large language models required significant GPU memory.
- Some models required authentication through a Hugging Face access token.
- Initial prompts produced inconsistent and repetitive responses.
- General-purpose models occasionally generated hallucinated medical information.
- Without proper prompt formatting, the chatbot sometimes repeated the user's question instead of answering it.

These challenges highlighted the importance of selecting the right model and designing effective prompts.

Improvements After Prompt Refinement

Prompt refinement significantly improved the chatbot's overall performance.

The following improvements were observed:

- Responses became more structured and easier to understand.
- Hallucinations were reduced by adding clear constraints.
- The chatbot followed instructions more accurately.
- Few-shot prompting improved the quality of structured summaries.
- Role-based prompts allowed the same model to generate responses appropriate for doctors, nurses, medical students, and patients.
- Overall response quality, readability, and reliability improved considerably.

Conclusion

This project demonstrated the practical use of Hugging Face Transformers and Prompt Engineering in developing an AI-powered healthcare assistant. By combining an instruction-tuned language model with well-designed prompts, the chatbot was able to generate structured, informative, and role-specific medical responses.

The project also highlighted that prompt engineering plays a crucial role in improving the accuracy, consistency, and safety of AI-generated content. Future improvements could include integrating medical knowledge bases, Retrieval-Augmented Generation (RAG), and fine-tuning the model using domain-specific healthcare datasets to further enhance performance.